Assignment 6

Logistic Distr:

$$\frac{d}{dx}\left(F(x;\sigma)\right) = \frac{d}{dx}\left(\frac{1}{1+e^{-x/\sigma}}\right)$$

$$f(x;\sigma) = \frac{d}{dx}\left(1+e^{-x/\sigma}\right)^{-1}$$

$$= -\left(1+e^{-x/\sigma}\right)^{-2} \cdot \frac{d}{dx}\left(1+e^{-x/\sigma}\right)$$

$$= -\left(1+e^{-x/\sigma}\right)^{-2} \cdot e^{-x/\sigma}\left(-\frac{1}{\sigma}\right)$$

$$= \frac{1}{\sigma}\left(\frac{e^{-x/\sigma}}{\left(1+e^{-x/\sigma}\right)^{2}}\right)$$

Exponential Distr: $F_X(x) = \begin{cases} 0, & x < 0 \\ 1-e^{-\lambda x}, & x \geq 0 \end{cases}, \quad \lambda > 0$

$$\frac{d}{dx}\left(1-e^{-\lambda x}\right) = \lambda e^{-\lambda x}, \quad x \geq 0$$

$$\frac{d}{dx}(0) = 0$$